

```
mindistance = distance
minidx = j
return(minidx)
```

[Listing 6-1:](#) *The `findnearest()` function, which finds the nearest city to a given city*

After we have this `findnearest()` function, we're ready to implement the nearest neighbor algorithm. Our goal is to create an itinerary called `nnitinerary`. We'll start by saying that the first city in `cities` is where our salesman starts:

```
nnitinerary = [0]
```

If our itinerary needs to have N cities, our goal is to iterate over all the numbers between 0 and $N - 1$, find for each of those numbers the nearest neighbor to the most recent city we visited, and append that city to our itinerary. We'll accomplish that with the function in [Listing 6-2](#), `donn()` (short for “do nearest neighbor”). It starts with the first city in `cities`, and at every step adds the closest city to the most recently added city until every city has been added to the itinerary.

```
def donn(cities,N):
    nnitinerary = [0]
    for j in range(0,N - 1):
        next = findnearest(cities,nnitinerary[len(nnitinerary) - 1],nnitinerary)
        nnitinerary.append(next)
    return(nnitinerary)
```

[Listing 6-2:](#) *A function that successively finds the nearest neighbor to each city and returns a complete itinerary*

We already have everything we need to check the performance of the nearest neighbor algorithm. First, we can plot the nearest neighbor itinerary:

```
plotitinerary(cities,donn(cities,N),'TSP - Nearest Neighbor','figure3')
```

[Figure 6-3](#) shows the result we get.